Mehmet Aygun

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Research Interest

I am interested in computer vision and machine learning. In particular, I am interested in developing image understanding models that can learn fundamental priors and representations through self-supervision alone. Examples of which include investigating the role of shape in semantic understanding.

Education

- **The University of Edinburgh (UoE), School of Informatics** — Edinburgh, UK
  Doctor of Philosophy in Computer Science, — *Sep. 2021 – Current*
- **Technical University of Munich (TUM), Department of Informatics** — Munich, Germany
  Master of Science in Computer Science — *Oct. 2018 – Dec. 2020*
- **Istanbul Technical University (ITU)** — Istanbul, Turkey
  Bachelor of Science in Computer Science — *Sep. 2013 – June. 2017*

Research Experience

- **Institute for Adaptive and Neural Computation at UoE** — Edinburgh, UK
  Postgraduate Research Student — *Sep 2021 - Current*
  - **Shape & Semantics**: Exploring the role of shape in semantic tasks like correspondence and fine-grained classification. Advisor: Dr. Oisin Mac Aodha
- **TUM Computer Vision Lab** — Munich, DE
  Research Student — *Oct 2018 - Jun 2021*
  - **Holistic Scene Analysis**: Worked on semantic/instance segmentation and multi object tracking with Lidar data. Advisors: Prof. Laura Leal-Taixe, Dr. Aljosa Osep.
  - **3D Shape Analysis**: Worked on shape correspondence problem with self-supervised deep learning methods using functional maps and heat kernels. Advisors: Prof. Daniel Cremers, Dr. Zorah Lahner.
- **ITU Computer Vision Lab** — Istanbul, Turkey
  Research Student — *Dec 2013 - June 2017*
  - **Medical Image Segmentation**: Worked on multi-modal 3D Brain Tumor segmentation using convolutional neural networks under supervision of Prof. Gozde Unal.
  - **Transfer Learning**: Worked on transfer learning for convolutional neural networks. Collaborated with Dr. Yusuf Aytar from MIT CSAIL and proposed a statistical regularization method for transferring knowledge between different CNN architectures.

Publications

- **M. Aygun**, O. Mac Aodha, ”Demystifying Unsupervised Semantic Correspondence Estimation”, In European Conference on Computer Vision (ECCV), 2022
- **M. Aygun**, A. Osep, M.Weber, M.Maximov, C. Stachniss, J. Behley, L. Lael-Taixe, ”4D Panoptic Lidar Segmentation”, In Conference on Computer Vision and Pattern Recognition (CVPR), 2021
- **M. Aygun**, Z. Lahner, D. Cremers, ”Unsupervised Dense Shape Correspondence using Heat Kernels”, In International Conference on 3D Vision (3DV), 2020
- **M. Aygun**, Y. Aytar, H. K. Ekenel, ”Exploiting Convolution Filter Patterns for Transfer Learning”, In ICCV TASK-CV workshop, 2017. **(Honourable Mention Award)**
- R. C. Malli*, **M. Aygun***, H. K. Ekenel, ”Apparent Age Estimation Using Ensemble of Deep Learning Models”, In CVPR Workshop, 2016 (*:equal contribution.)

Honors & Awards

- **ICCV 17 Workshop, Honourable Mention Award**: Received Honourable Mention Award at ICCV 17 TASK-CV workshop for our paper "Exploiting Convolution Filter Patterns for Transfer Learning".
- **DAAD scholarship**: Two years of master scholarship for studying in Germany.
- **Bachelor Thesis Award**: Best Bachelor Thesis Award from Computer Engineering Department for my thesis titled "Multi Modal 3D Convolutional Neural Networks for Brain Tumor Segmentation"
- **Undergraduate Research Scholarship** : Research scholarship from science and technology research council of Turkey during my last two year of bachelor education.

Professional Experience

- **Coriolis Pharma** Munich, Germany
  Computer Vision Engineer *Sep 2019 - March 2020*
  - **Biological image analysis**: Developed methods for particle analysis using deep learning based classification, instance segmentation and unsupervised learning methods. Used frameworks: Tensorflow, Keras, OpenCV.
- **Motius** Munich, Germany
  Machine Learning Engineer *Jan 2019 - April 2019*
  - **Handwritten Text Recognition**: Developed handwritten text recognition system using deep learning (CNN+LSTM+CTC). Reduced word error rate from 18% to 12%. Used frameworks: Tensorflow, OpenCV.
- **Bunsar** Istanbul, Turkey
  Computer Vision Engineer *Feb 2017 - Sep 2018*
  - **Visual Search Engine from Images and Videos**: Research and development activities for image retrieval system using deep learning methods. Topic studied: Object Detection, Image Segmentation, Image Classification, Large Scale Image Retrieval Used frameworks: Caffe, OpenCV, Sklearn

Skills

- Python, C/C++, PyTorch, OpenCV, Numpy, Matlab, Linux, Git

Related Coursework

- **Technical University of Munich**: Natural Language Processing, Introduction to Deep Learning, Variational Methods for Computer Vision, Multiple View Geometry, Tracking and Detection
- **Istanbul Technical University**: Introduction to Computer Vision, Learning from Data, Probability, Linear Algebra, Calculus I & II

Activities

- Reviewer for CVPR22, ECCV22, 3DV22, FGVC9
- Tutoring for Introduction to ML Course UoE 2021
- Volunteering work at code.org for high school students between 2015-2017

References

- **Dr. Oisin Mac Aodha**: oisin.macaodha@ed.ac.uk
- **Prof. Laura Leal-Taixe**: leal.taixe@tum.de
- **Prof. Gozde Unal**: gozde.unal@itu.edu.tr